

TEAM ID	SWTID-2026-7349
PROJECT TITLE	India's Agricultural Crop Production Analysis using Tableau
MAX MARKS	1 MARK

CHAPTER – V

PROJECT PLANNING AND SCHEDULING

5.1 – Project Milestones & Tasks

1. Data collection: We get data from government websites like the Ministry of Agriculture portal and other good sources. We look at numbers like how much crop is produced how much land is used how much crop is grown per unit of land, rainfall, temperature and fertilizer use in different parts of India like states, districts and seasons. We make sure we have data from years like 1997 to 2014 so we can see how things change over time.

2. Data Preparation: We clean up the data by removing duplicates fixing missing information and making sure names of crops and states are the same everywhere. We change the data from a list to a wide table so we can look at it over time and deal with problems like districts changing over time. We put the data together by state, season, type of crop and year so it is ready to be looked at and shown in pictures.

3. Data visualization: We make pictures and charts in Tableau to see how agricultural production is changing. We draw lines to show how production changes over time make charts to compare seasons like Kharif and Rabi crops and maps to show what is happening in each state. We also make scatter plots to see how good different crops are. We look for things like when production is highest which states are doing best and what happens at different times of the year.

4. Dashboard: We put all the pictures and charts together in one place so people can look at the data in ways. We add filters, buttons and other tools so people can easily look at the data for states, seasons and types of crops. We show numbers like total production, total land used and how many crops there are so people can see them quickly.

5. Story: We show the important things we found out in a way that tells a story with the data. We put the pictures and charts in order with text that explains what is happening, like trends, surprises and what is happening in different parts of the country. We give people who make decisions like government officials, farmers and agricultural businesses information they can use to make choices.

6. Performance testing: We make sure the dashboard opens quickly and works smoothly when people use it. We use a tool called Tableaus Performance Recorder to see how long it takes to get the data and show the pictures. We find out what is slowing it down and make it better by using techniques like simplifying the data and calculations.

7. Web Integration: We make the Tableau dashboard work on the web so people can see it from anywhere. We use codes and tools like the Tableau Embedding API to put it on websites and web applications. We make sure only the right people can see it by using keys and passwords.

8. Project demonstration & documentation: We show the project to the people who are interested and give them a live tour of the project and its features. We write down everything, about the project like how to use it what the data means and how it works so people can understand it. Use it in the future. We make sure everything is written down so people can learn from it and use it to make decisions.